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TITLE: AI SaaS Platform: Integrated OpenAI APIs for Text Generation, Image Generation, and Resume Scoring
SCHOOL: Chitkara University, Rajpura, Punjab

DETAIL OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT PORTAL — NEED TO SHARE LOGIN AND PASSWORD WITH MENTORS ALL MEMBERS DETAILS INCLUDING SUPERVISOR

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ABSTRACT

The AI SaaS Platform is designed to provide a centralized and scalable solution for delivering multiple AI-powered services through a single unified platform. In today's digital environment, users often rely on different AI tools for tasks such as text generation, image creation, resume analysis, and workflow automation. These tools are usually distributed across multiple platforms, resulting in fragmented user experiences, inefficient workflows, higher costs, and complex management processes.

The proposed platform addresses these challenges by integrating various AI functionalities into one secure and user-friendly ecosystem. The system offers services such as AI-based text generation,

image generation, resume scoring, smart content assistance, and automated productivity tools through an intuitive web interface. By leveraging OpenAI APIs and modern cloud technologies, the platform enables users to access advanced artificial intelligence capabilities efficiently without switching between multiple applications.

The platform is developed using a modern full-stack architecture consisting of React.js and Tailwind CSS for the frontend, Node.js with Express.js for backend services, and MongoDB for secure and scalable data management. Authentication and user management are handled using Clerk, while Stripe integration provides secure subscription and payment processing.

The system also includes role-based access control, subscription-based plans, real-time analytics, and user activity tracking to ensure scalability, security, and better resource management. The modular architecture allows easy integration of future AI services and supports high scalability for growing user demands.

This AI SaaS Platform aims to simplify access to artificial intelligence solutions, improve productivity, reduce operational complexity, and provide a seamless digital experience for individuals, students, freelancers, and businesses. The project demonstrates how modern AI technologies can be integrated into a scalable SaaS ecosystem to deliver intelligent, efficient, and user-centric solutions.

1.1 Problem Background

With the increasing demand for AI tools, users often rely on multiple platforms for different functionalities like content generation, image creation, and resume evaluation. This fragmented ecosystem leads to:

- **Inefficient Workflows:** Users need to switch between multiple AI platforms for tasks like content generation, image creation, and resume analysis. This consumes time, interrupts workflow, and reduces overall productivity.
- **Lack of Centralized Access:** Most AI services operate independently, making it difficult for users to manage data, projects, and outputs from a single place. A unified system is missing.
- **Poor User Experience:** Different platforms have different interfaces, login methods, and pricing structures, which creates confusion and makes the overall experience less smooth and user-friendly.
- **Limited Scalability and Security:** Standalone AI tools may not efficiently handle increasing users and data. Managing sensitive information across multiple platforms can also create privacy and security concerns.
- **Need for a Unified Platform:** There is a need for a centralized AI SaaS platform that integrates multiple AI-powered services into one secure, scalable, and user-friendly ecosystem for better efficiency and management.

1.2 Objective of the Project

The main objectives of this project are:

- To build a centralized AI SaaS platform integrating multiple AI functionalities
- To provide text generation, image generation, and resume scoring in one system
- To implement secure authentication and user management
- To enable subscription-based access using a payment gateway
- To ensure scalability and performance using modern web technologies

1.3 Proposed Solution / System Overview

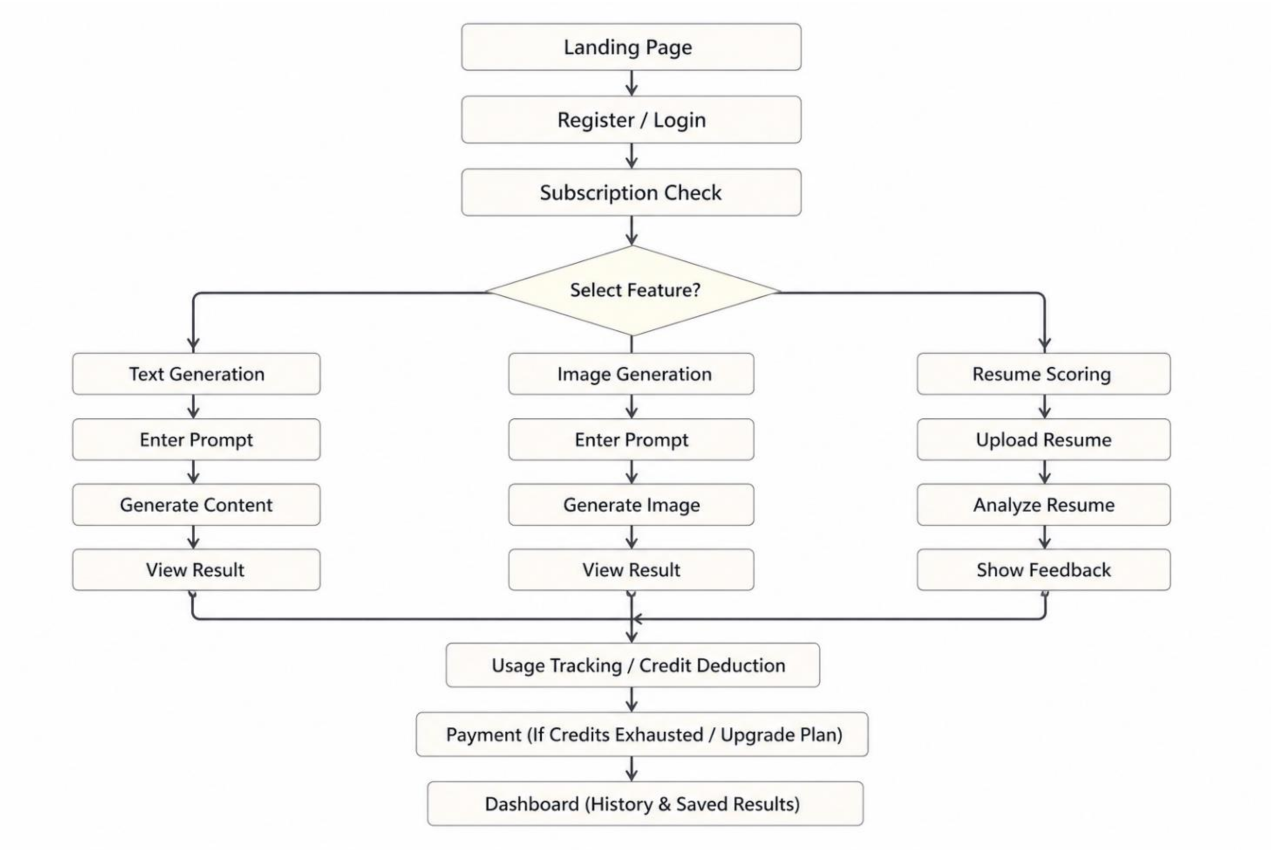
The AI SaaS Platform is a full-stack web application built using the PERN stack — PostgreSQL, Express.js, React.js, and Node.js. This technology stack was chosen to ensure a scalable, secure, and high-performance system while maintaining consistency across the application using JavaScript for both frontend and backend development.

The frontend is developed using React.js, providing users with a fast, responsive, and seamless experience without frequent page reloads. The backend runs on Node.js with Express.js handling all API routes, business logic, and integration with external AI services. PostgreSQL is used as the database to store structured user data such as account details, subscription information, generated content, and usage records efficiently.

For authentication and security, JWT and bcrypt are implemented to manage user login, session handling, and data protection. OpenAI APIs are integrated to enable core functionalities like text generation, image generation, and resume scoring. Stripe is used for managing subscription-based payments, ensuring secure and reliable transactions along with automated billing workflows.

Additionally, the platform is deployed using modern cloud services, with the backend hosted on Render and the frontend on Vercel, ensuring scalability, reliability, and optimized performance for users.

1.3.1 Platform Workflow



1.4 Key Features and Functionality

The AI SaaS Platform is built around the following features:

● Secure Authentication & Access Control:

The platform uses JWT and bcrypt for secure login, user management, and controlled access to features based on subscription plans.

● AI-Powered Content & Image Generation:

Users can generate text content and images by providing prompts. The system uses OpenAI APIs to deliver fast and accurate results within a single interface.

● Resume Scoring and Feedback:

Users can upload resumes, and the platform analyzes them using AI to provide scores and improvement suggestions.

● Subscription & Usage Management:

Stripe is used for handling payments and subscriptions, while the system tracks usage and provides a dashboard to manage generated results.

1.5 Technology / Method Used

Technology	Purpose
React.js	React is used as the frontend library to build a responsive and interactive user interface. It provides reusable UI components for features like text generation, image generation and dashboard.
Node.js + Express.js	Express is used for building the backend where API routes, business logic, authentication, and integration with AI services are handled efficiently.
PostgreSQL	PostgreSQL is used to store user information, subscription details, generated results, usage records and payment history in a structured and reliable format.
JWT + bcrypt	JWT is used for secure authentication and **bcrypt** is used for password hashing to ensure safe login, authorization and session management.
OpenAI APIs	OpenAI APIs power the core AI features including text generation, image generation and resume scoring & feedback.
Stripe	Stripe is integrated for handling subscription plans, secure payments, invoices and transaction management.

1.6 Expected Outcome and Benefits

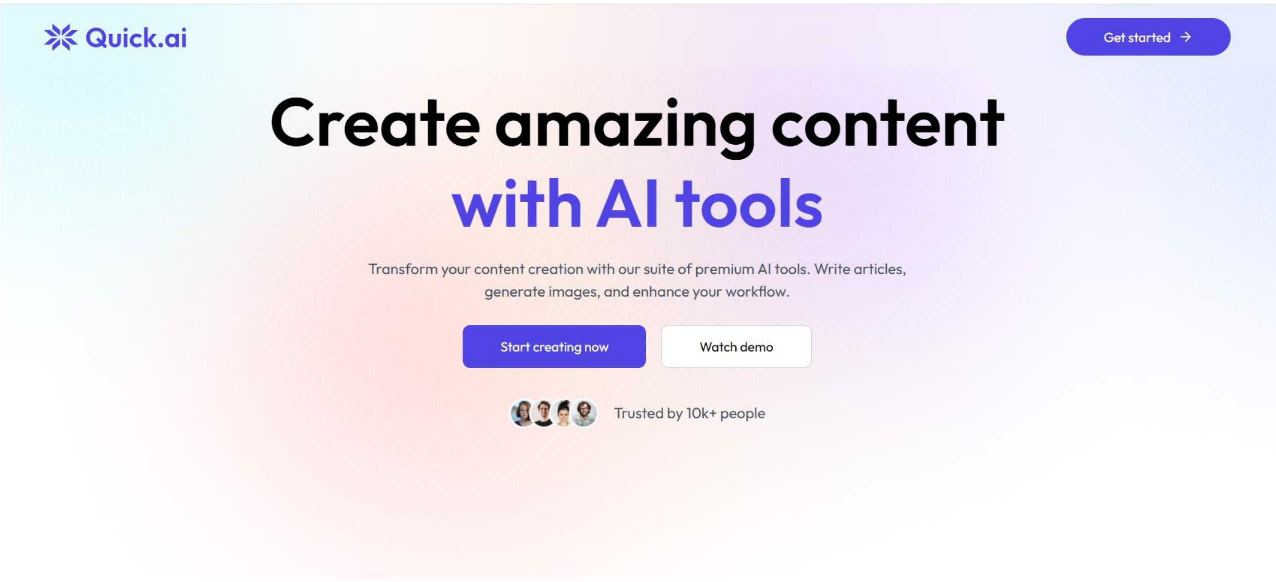
Post deployment, the AI SaaS Platform is expected to significantly simplify access to multiple AI-powered tools within a single system for users.

For users, it eliminates the need to switch between different platforms for tasks like content writing, image generation, and resume evaluation. All features are integrated into one unified interface, providing instant results and a smooth, efficient workflow with minimal effort.

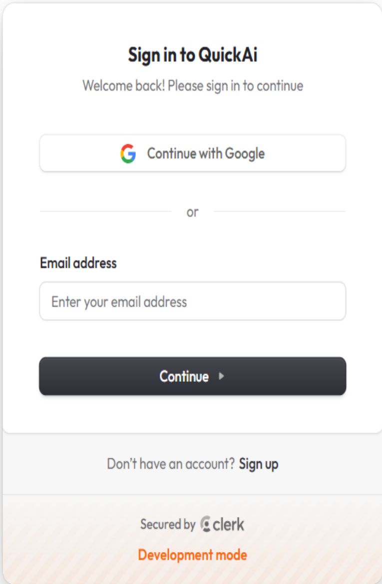
The platform also ensures secure authentication, reliable performance, and structured usage through subscription-based access. Users can easily manage their activities, track usage, and access previously generated results through a centralized dashboard.

Overall, the AI SaaS Platform aims to provide a modern, efficient, and scalable solution that enhances productivity by bringing multiple AI capabilities together in a fast, secure, and user-friendly environment.

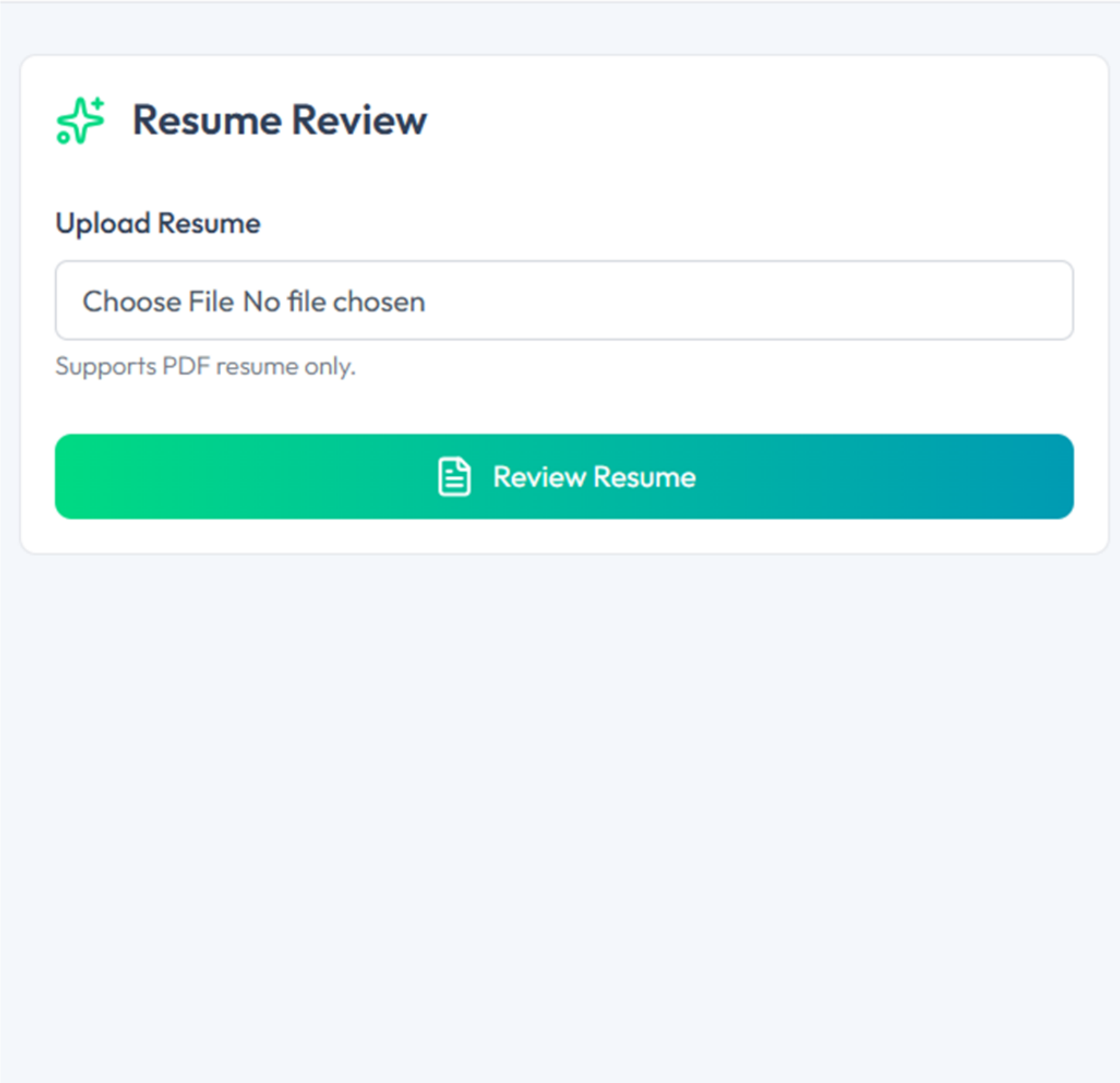
1.6.1 Landing Page



1.6.2 Login using Clerk



1.6.3 Resume Reviewing using AI



1.7 Original Contribution Statement

The AI SaaS Platform was developed entirely by me as part of my academic project at Chitkara University, Rajpura, Punjab, under the guidance of Dr. Swati Goel. Throughout the development process, Dr. Swati Goel provided guidance and feedback whenever necessary, while all design decisions, implementation, and documentation were carried out independently.

The platform has been built from scratch without using any pre-built templates, borrowed codebases, or copied logic from existing projects. All third-party tools and services used, including OpenAI APIs, Stripe, and other libraries, have been integrated in accordance with their respective licensing terms.

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